

OSI Model

* OSI model stands for Open Systems Interconnection Model) is essential model used in networking. This critical Model provides a framework dictating how all networked devices will send, receive and Interpret data.

* One of main benefits of OSI model is devices can have different functions and designs of network while communicating with other devices (Ex! - iPhone, Smart TV, gaming PC, they all look different, run different software while communicating with each other device your iPhone can send email to windows PC without any formatting issues), Data sent across a network that follows uniformity of OSI model can be understood by other devices.

* The OSI has 7 layers. Each layer has different set of responsibilities and arranged from layer 7 to layer 1. This process is called encapsulation.

7.	Application
6.	Presentation
5.	Session
4.	Transport
3.	Network
2.	Data link
1.	Physical

Quiz:-

1) What does "OSI" in "OSI Model" stand for?

A) Open Systems Interconnection.

2) How many layers (In digits) does OSI model have?

A) 7

3) What is key term when pieces of information get added to data?

A) Encapsulation.

Layer 1:- Physical :- This layer is easiest layer to grasp. This layer references the physical components of hardware used in networking and is lowest layer that you will find.

Devices use electrical signals to transfer data between each other in binary numbers.

system (1's and 0's) .

Ex!:- ethernet cables connect devices .

Quiz:-

1) what is the name of this layer? | 2) what is the name of numbering system that is both 0's and 1's?
A) Physical | A) Binary

3) what is name of cables that are connected to other devices?

A) Ethernet cables

2. **Data link** : — The data link layer focuses on physical addressing of transmission. It receives a packet from Network layer (including IP Address) and adds in Physical MAC (Media Access Control) address for receiving endpoint .

* Inside every network enabled computer is a Network Interface card (NIC) which comes unique MAC address to identify it .

* MAC addresses are set by manufacturer and literally burnt into the card, they can't be changed. Although they can be spoofed. When information sent across network, its actually physical address that is used to identify whom exactly to send the information .

The job of data link layer to present data in format suitable for transmission .

Quiz:-

1) what is name of this layer? | 2) what is name of piece of hardware that all networked devices comes with?
A) Data link | A) Network interface card

3) **Network layer** : — The 3rd layer of OSI model (network layer) where magic of routing and reassembly of data takes place (from small chunks to larger chunks).

→ Firstly routing simply determines most optimal path in which these chunks of data should be sent .

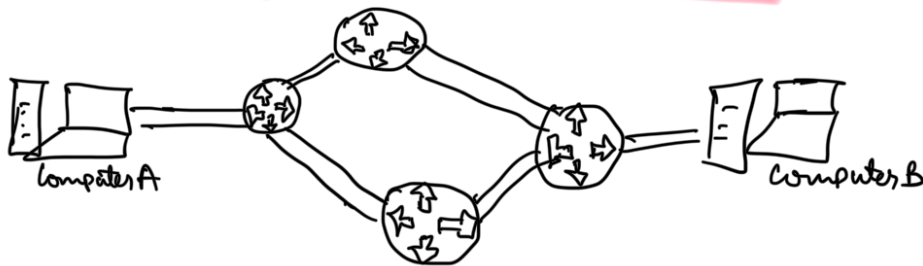
* Some protocols at this layer determine exactly optimal path that data should take to reach

the device.

* Briefly these protocols include OSPF (Open Shortest Path First) and RIP (Routing Information Protocol). The factors that decide what route is taken is decided by

- 1) What path is shortest? 2) What path is most reliable 3) ^{which path has faster} Physical connection?

At this layer Everything is dealt with IP Addresses such as 192.168.1.100. Devices such as router capable of delivering packets using IP Addresses are known as layer 3 devices.



Quiz:-

1) What is the name of this layer? 2) Will packets take most optimal route across a network (Y/N)?

A) Network

A) Y

3) What does the acronym "OSPF" stand for?

4) What does acronym "RIP" stand for?

A) Open Shortest Path First

A) Routing Information Protocol

5) What type of addresses are dealt with at this layer?

A) IP addresses